

Revitalizing Non-Motorized Transportation in Policy and Built Environment: A Bike Prohibition Case Study of Wujiaochang, Shanghai

Abstract: Bicycle transport is crucial for low-carbon living, urban health, and addressing the last-mile issue. However, urban motorization has eroded non-motorized rights of way in China, once known as the 'kingdom of bicycles'. Policies have further limited non-motorised spaces amidst rising demand due to the sharing economy and carbon reduction goals. Land constraints also hinder road expansion, escalating conflicts between traffic modes. This study assesses the 2005 'no non-motorised vehicle' policy in Shanghai's Wujiaochang area. Despite some roads had been reopened to cycling in 2021, about 100km remain inaccessible to non-motorised traffic. Secondary roads have assumed the role of diverting non-motorised traffic, but their cycling environment's effectiveness and safety are uncertain. Using sensor and Street View technologies, the study introduces practical indicators to evaluate secondary roads' cycling environments, including road smoothness, light intensity, and spatial features. Findings indicate these roads are ill-equipped to meet cycling safety and efficiency needs. The study advocates for preserving non-motorised rights of way to foster low-carbon development and urban economic vitality. Objective physical indicators are proposed to measure non-motorised roadway utility, guiding future policy adjustments. These aim to enhance cycling environments, ensure cyclists' equitable road access, and promote transport equity. The study encourages a shift towards a sustainable, people-centred urban model, aligning with the well-being of its inhabitants.

Keywords: Bicycle transportation, Prohibition of non-motorized vehicles, Non-motorized utility indicators, Sustainable development

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1. Introduction

The prohibition on cycling is a measure enacted to uphold traffic order and ensure the safety of pedestrians, cyclists, and motorists by restricting cycling in designated areas. Shanghai's government reports that currently, 107 roads are off-limits to non-motorised vehicles, a policy that reflects the city's dynamic urban traffic context.

This initiative stems from the need to address the escalating issue of urban congestion, exacerbated by the city's expansion and a proliferation of private vehicles. As part of the solution, Shanghai introduced the 'Separation of Motorised and Non-Motorised Traffic' in 2004, which has led to the exclusion of bicycles from certain city roads. Additionally, the cycling ban is justified on safety grounds, given the dense motorised traffic and intricate traffic patterns that could pose heightened risks to cyclists.(汤, 刘 和 潘, 2014)

However, this approach has faced criticism, as evidenced by the case of Frankfurt, where the merereallocation of cyclists to side roads without improved infrastructure has resulted in increased accidents and ongoing safety concerns for those crossing major roads. (熊, 陈 和 胡, 2010)This scenario prompts a critical evaluation: Is the no-cycling policy well-suited to the current urban traffic landscape? Does it effectively promote the objectives of a well-developed slow-moving traffic system?

1.1 The Importance of Riding

Urbanization is accelerating, and so is the development and size increase of motor vehicles. Yet, in recent years, a growing ecological consciousness among the populace has been a notable shift. Alongside the chronic issues of traffic congestion, parking challenges, and vehicular air pollution, there's a burgeoning trend: an increasing number of urban dwellers are selecting bicycles and electric bicycles for short-distance travel. The advent of bike-sharing systems has further bolstered the adoption of bicycles as a sustainable alternative to motor vehicles for such journeys. Bicycles offer several comparative advantages, including ecological sustainability, enhanced speed and convenience, health benefits through exercise, equitable accessibility for residents, and a boost to the sharing economy.

Bicycles play an important role in efforts to create sustainable and just societies of the future by reducing carbon dioxide emissions and challenging car dependency, are virtually silent and pollution-free, take up a smaller portion of road and parking space than cars(Pucher and Buehler, 2017), increase social well-being over short distances, and enrich the transport choices of disadvantaged groups such as low-income earners, the elderly and children(Schwanen and Nixon, 2020). Furthermore, bicycles require fewer resources to produce and maintain, create new demand for new infrastructure and urban spaces(Henriksson, Wallsten and Ihlström, 2022), and sharing bicycles rather than privately owning them reduces the use of energy and raw materials and is cheaper for the individual user as the cost of renting them can be cheaper than purchasing them(Sochor, Strömberg and Karlsson, 2015). Shared communities can also create local social networks and support social sustainability. In conclusion, bike sharing as a green alternative to motorised travel can contribute to a sustainable and more circular future.In conclusion, bike sharing as a green alternative to motorised travel can contribute to a sustainable and more circular future.

Media attention has led to a surge in social awareness and the cultivation of a bicycle culture. An increasing number of individuals are drawn to cycling for the sense of agency it provides. Others are motivated by a preference for cycling in their local areas, potentially due to the appeal of mixed land uses in their vicinity(Xing, Handy and Mokhtarian, 2010). Since cycling is largely powered by the cyclists, it offers

an excellent opportunity for exercise, beneficial for cardiovascular health and for tackling physical health issues, a fact that is gaining wider acceptance. In the modern societal context, cycling is instrumental in the concept of "rebuilding the neighborhood," encouraging a renewed focus on the immediate surroundings within a 1.5km radius.

In conclusion, the practice of cycling extends beyond personal health benefits and environmental conservation to being intricately tied with social progress and urban planning. Recognizing its multifaceted importance, the need for the development and execution of cycling policies has become self-evident.

1.2 History of cycling policy development

Cycling policies have evolved globally from discouragement to active promotion, reflecting a shift from car-centric to multimodal transport planning. Despite the dominance of automobiles since the industrial revolution, even cycling-friendly nations like the Netherlands, Germany, and Denmark experienced a decline in cycling from 1950 to 1975. It was the mid-1970s that marked a significant policy reversal, leading to the current cycling renaissance. Government policies have been pivotal in this transformation, outweighing individual experiences in shaping cycling's trajectory.

Since the 1980s, European cycling policy has matured, with the Netherlands, Germany, and Denmark providing comprehensive support, including objectives, guidelines, funding, and infrastructure development to promote cycling. These countries have established national master plans aimed at increasing daily cycling and enhancing safety through improved lane design, increased parking, public transport integration, and safety measures.

The European Union (EU) has played a significant role in fostering cycling through initiatives like the Interreg program, which funds cross-border cycleway projects, complementing the European Cyclist Federation's (ECF) VeloEuro program. The EU also supports research and knowledge exchange on cycling best practices.

Shanghai's cycling policy has transitioned through two distinct phases. The initial phase, marked by the 2002 White Paper on Urban Transport, aimed to reduce bicycle numbers by 25% and proposed converting non-motorised lanes into public transport or bus-only lanes. However, the 2013 White Paper shifted focus, prioritising the development of slow-moving transport systems, including cycling and walking. Policies were introduced to improve cycling infrastructure, regulate illegal parking, and develop public bicycle rental systems. Despite these efforts, the ban on bicycles in certain areas was seen as outdated.

In 2021, Shanghai issued the "Guidelines for the Planning and Design of Slow-moving Transport in Shanghai," advocating for a dense road network and prioritising the development of slow-moving systems. The 2022 White Paper on Urban Transport further emphasized the creation of an inclusive and vibrant slow-moving transport network, with a focus on protecting the right-of-way for these modes of transport. The "Three-Year Action Plan on Accelerating the Improvement of the Quality of Slow-moving Transportation in the City (2023-2025)" issued by the Shanghai Municipal Transportation Commission outlines plans to address network bottlenecks, restore access on previously restricted roads, and explore the creation of non-motorised lanes to complete the slow-moving transport network.

1.3 Reasons for cycling prohibition policy

Cycling prohibition policies are fundamentally driven by spatial and temporal conflicts between motorized and non-motorized vehicles in urban development, and a history of motorized-oriented planning has also influenced their enactment. Literature shows that removing non-motorized vehicles from the road is intended to reduce motor-non-motorized conflicts, although there is no corresponding data to support this claim.

The Wujiaochang roundabout area in Shanghai's Yangpu District was selected for this study. Since the Central Ring Road's opening in 2005, the public security traffic management department has implemented measures prohibiting non-motorized vehicles from passing through some or all of the near roundabouts of five arterial roads: Handan Road, Siping Road, Huangxing Road, Xiangyin Road, and Songhu Road. Concurrently, five major shopping malls—Wanda Plaza, Bailian Yiyicheng, Yumai Lifestyle Plaza, Hesheng Hui, and Suning City Plaza—have been completed and opened, formally establishing the area's "five corners." The enhancement of commercial services has increased traffic pressure, with rising traffic flows and complex interactions between motor vehicles, non-motorized vehicles, pedestrians, and public transport. Given these conditions, the introduction of the no-cycling policy was a product of its time and historical context. However, it is pertinent to question whether this policy remains suitable for current traffic development needs.

1.4 Survey of cycling prohibition policy

For this reason, we distributed questionnaires around the Wujiaochang roundabout, with a total of 121 sets distributed, of which 113 sets were valid. We inquired about the no-riding policy in the Wujiaochang roundabout area (Figure 1). Among the interviewees, 54.69% were completely unaware of the no-riding policy, and only 14.06% had a good understanding of it. The majority of respondents reported traveling through the area less frequently due to the policy. Notably, the same proportion of respondents (14.06%) had been fined under the no-riding policy. The no-riding policy's popularity in Wujiaochang is low, with over half of the respondents having received fines and considering the policy's impact on their lives to be significantly negative. Furthermore, the policy's popularity is insufficient among the interviewees, with a high number unaware of it. The proportion of those with a high level of understanding correlates with the number of people who have been fined.

When examining respondents' opinions on specific no-riding road sections, most felt that the no-riding policy on Handan Road (from Yixian Road to Guoding Road) was unreasonable, while the policy around the Wujiaochang roundabout was deemed reasonable. However, the policy on Xiangyin Road (from Guoho Road to Jungong Road) was considered unreasonable. Regarding the capacity of roads adjacent to no-riding sections (Guoding Road, Anbo Road, Zhengtong Road) to accommodate non-motorized traffic, over half of the respondents believe these roads are incapable of handling the volume of non-motorized trips and that improvements to the riding environment are necessary.

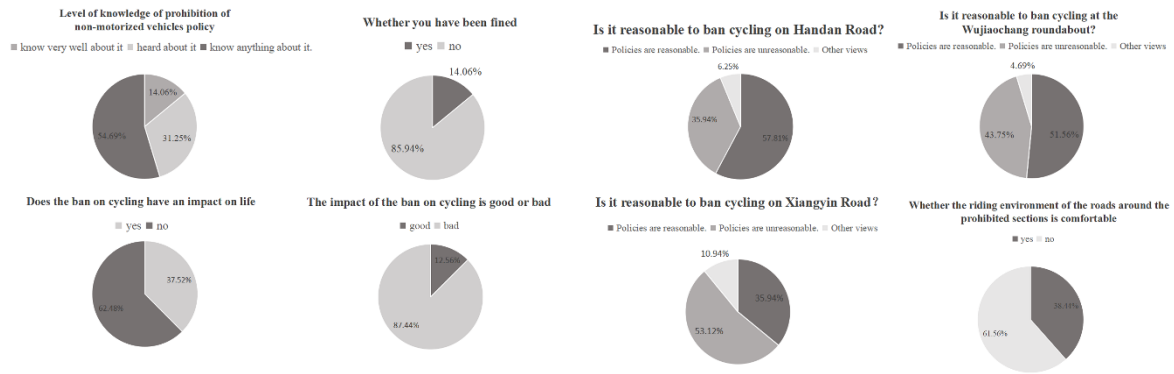


Fig.1 Investigation results of the prohibition of non-motorised vehicles policies

To date, after the irrationality of the no-riding policy was recognised, government administrators have taken some measures. For example, some of the banned sections have been lifted: in Jungong Road (Xiang Yin Road - Song Hua Jiang Road), a non-motorised lane with a width of about 2 metres has been left out of the 4-metre pavement space through the design of a common slab for both people and non-motorised vehicles, with the use of different paving materials or street trees to separate the space for both people and non-motorised vehicles. Propose traffic improvement measures on Handan Road (Guoshun Road - Guonian Road intersection) by setting up a special non-motorised crossing mainly for Fudan University students who go to school across the two campuses. Alternatively, the cycling environment of the roads around the no-cycling area could be enhanced, for example, by improving the road section of Zhengtong Road so that the motorway, non-motorway and footpath are not at the same height. The footpath is located at the upper street edge of the pad, which is at the highest ground level, followed by the non-motorised carriageway and the carriageway at the lowest level. However, a large number of illegally parked takeaway vehicles still encroach on the non-motorised carriageway and footway space during the peak lunchtime and evening food distribution periods. But this by encroaching on the walking space to make way for non-motorised cycling behaviour, not only did not address the root causes of motor vehicles and non-motorised road right-of-way allocation, but also encroached on the right-of-way of the walker, the machine non-human contradiction still exists and more prominent.

1.5 Purpose of the thesis

Since the implementation of the no-cycling policy in Shanghai, there has been considerable negative feedback. In response, the government has become more proactive in mitigating the negative impacts of the ban. A decade after the ban, it is pertinent to assess whether the roads surrounding the restricted areas can accommodate the increased non-motorised traffic flow. To this end, we have defined a set of evaluation indicators to measure the comfort of the cycling environment on neighboring roads. These indicators include motorised and non-motorised segmentation, road smoothness, road width, green visibility (a streetscape parameter), the skyline index, and the ratio of buildings along the street. We have also categorized the study period into daytime and nighttime assessments. Through these indicators, we aim to evaluate the capacity of surrounding roads to handle the traffic flow diverted from the no-riding streets. Additionally, we will assess the effectiveness of Shanghai's no-riding policy by determining the safety, comfort, and convenience of riding on these alternative streets. This will help identify any shortcomings and establish a comprehensive set of evaluation indicators. These indicators will serve as a benchmark and guide for

transforming the riding environment. Ultimately, we will provide recommendations for adjusting the direction of Shanghai's no-riding policy and for enhancing the environment of the streets surrounding the no-riding zones.

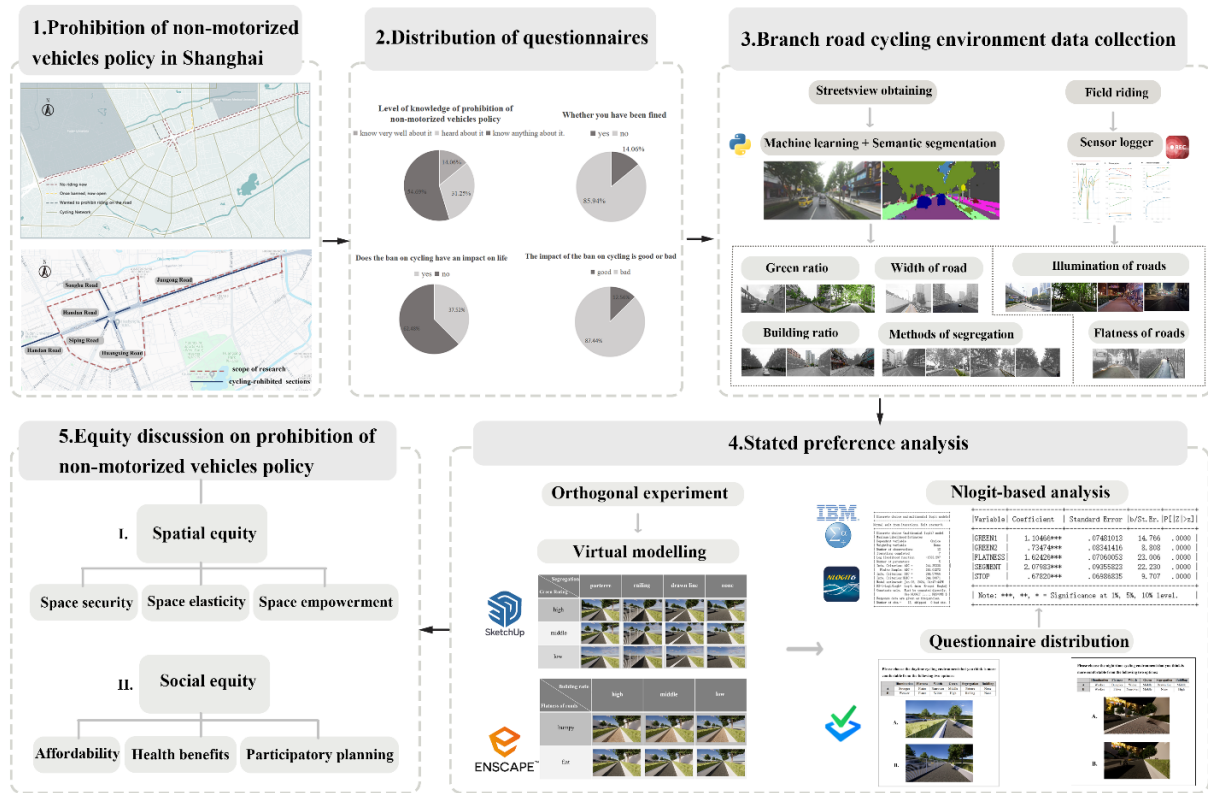


Fig.2 Technical path

2. Methodology

2.1 Technical background

Studies assessing cycling ability usually calculate various indicators that shape good cycling conditions and combine them into quantitative indices for comprehensive assessment, which can be broadly categorised into three types of indicators: network-based, environmental, and evaluative. Network indicators are mostly used to express the connectivity of cycling roads, which influences cyclists' route preference (Dill, 2004; Buehler and Dill, 2016). For example, the degree of integration and choice in spatial syntax, or the continuity and density of the road itself. Most studies have shown a positive correlation between bikeway networks or aspects of the network and levels of cycling (Buehler and Dill, 2016), where road discontinuities are prone to serious safety issues (Nabavi Niaki, Saunier and Miranda-Moreno, 2019). Environmental indicators are mostly used to describe specific cycling spaces, such as the degree of greenery, light intensity, spatial form of streets, lane slope width, non-motorised zones, land use, noise, buildings, water bodies, pedestrians, vehicles, etc. (Rodríguez and Joo, 2004; Sener, Eluru and Bhat, 2009; Majumdar and Mitra, 2017). Green elements have been widely shown to enhance the comfort and attractiveness of urban streetscapes (Lu, Sarkar and Xiao, 2018), and the positive effects of street tree greenery on cycling have been well explored (Nawrath,

Kowarik and Fischer, 2019). Lighting issues, i.e. light intensity, are an important factor influencing cycling behaviour at night (Poorfakhraei and Rowangould, 2015), and good lighting at night improves cyclists' visibility and makes it easier to identify road conditions and signs, even in poor weather conditions. Good lighting at night will improve the visibility of cyclists, make it easier to identify road conditions and signs, and keep them warm even in poor weather conditions (Gatersleben and Appleton, 2007; Poorfakhraei and Rowangould, 2015). And good daytime lighting helps to alleviate visual fatigue and eye stress and guide safe route choices. Steeper grades in lanes increase the difficulty and safety risks of cycling, while wide, flat lanes provide greater access and safety (Horacek *et al.*, 2012; Koh and Wong, 2013), and mechanically undivided strips reduce motor vehicle traffic conflicts and improve cycling safety and convenience. Evaluation-type indicators tend to refer to factors which are difficult to be measured in relation to the comprehensive perception of cyclists, such as the attractiveness, safety, and tidiness of the street (Clifton, Livi Smith and Rodriguez, 2007). In an experiment on the quality of public transport services measured by some scholars, the level of risk was found to be the most important factor influencing cycling choice (Majumdar and Mitra, 2017), in the streetscape greening study, the enhancement of street attractiveness by green elements had a significant effect on cycling (Nawrath, Kowarik and Fischer, 2019; Rita *et al.*, 2023).

Street view imagery (SVI) refers to images that depict urban landscapes in a similar way to human perspectives, and it has become an important data source for urban spatial representation and urban analysis in the past research city (Biljecki and Ito, 2021). Compared to traditional data sources, SVI has the advantages of high coverage, large data volume, relatively low data bias, cost-effectiveness and low time (Rzotkiewicz *et al.*, 2018; Gage *et al.*, 2019; Biljecki and Ito, 2021). Combined with the current advances in machine learning and computer vision technologies, object detection (Pathak, Pandey and Rautaray, 2018), semantic segmentation (Wang *et al.*, 2018), image classification (Rawat and Wang, 2017) and other technologies, which help to enhance the machine's understanding of the image, extract high-dimensional deep information, and provide a solid foundation for the observation of the urban physical environment (Kang *et al.*, 2020). Growing computer arithmetic power makes it possible to automate the processing of massive amounts of urban information (Biljecki and Ito, 2021).

Among the many methods for determining indicator weights, unlike surveys of real-life residential environmental elements, the method of deriving the relative importance of environmental elements through preferences for virtual scenarios is more advantageous, known as Stated Preference, or SP for short. SP is a method of estimating indicator weight coefficients through modelling, and consists of a number of logit models (Prioni and Hensher, 2000; Hensher, 2001; Hensher and Prioni, 2002). Logit models are based on the discrete choice model of random utility theory, where the probability of choosing each option is calculated. The utility of each choice is a combination of systematic and random components. All stochastic components are distributed according to Gumbel random variables. The estimation of the model coefficients allows assessing the importance of different indicators for the dependent variable. Among the mixed logit models (Hensher, 2001) the SP data compared to the display preference (RP) data, SP can take into account a wider attribute space, i.e., observed or unobserved influence dimensions, than RP data. Models based on SP surveys are well established and suitable for assessing attributes or estimating WTP, especially in the context of cycling (Poorfakhraei and Rowangould, 2015). However, the use of SP methods for residential environmental quality assessment in China has just started. This study attempts to establish a complete evaluation framework for cycling roads by building a ML model to evaluate the relative magnitude of the utility of each cycling indicator.

2.2 Experimental pathways

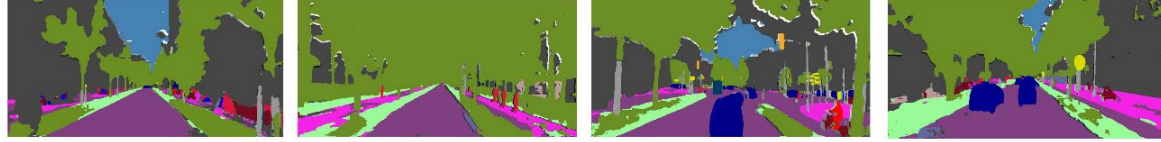
Baidu Maps is a web mapping service provided by Baidu, offering panoramic photos that include the vast majority of urban spaces in China. Since 2013, it has been providing a street view service, Baidu Panorama. In order to measure several specific cycling metrics, this study collected 46 pieces of street scene data from the study section of Wujiaochang in February 2021, with a total length of about 7.8km, and crawled a street scene photo for about 150m, combined with PSPnet semantic segmentation technique(Yang *et al.*, 2018). Through the recognition of trees, buildings, streets and sky, the four physical attributes of the street, namely green visibility, the percentage of buildings along the street, the road width, and the mechanical and non-physical separation, are calculated(Figure 3). In addition, in order to further quantify the illuminance factor and road flatness, which affect cycling choices, the researchers carried out field recordings and measurements using a mobile phone app, SensorLogger, which is an all-in-one sensor that integrates accelerometers, gyroscopes, GPS, and photometers into a single unit on a mobile phone. SensorLogger is a sensor that integrates accelerometer, gyroscope, GPS, photometer and other functions on mobile phone, which is usually used for data analysis, research or monitoring. In this experiment, light intensity and vertical acceleration z were used to measure the illumination and flatness of the riding road. In addition, two sets of light intensity data were collected, one for daytime natural light and one for nighttime light, in order to explore the difference between morning and evening riding preferences.

In the experiment of SP method to define factor weights, we firstly adopt the natural intermittent point grading method to determine the research level for the basic data collected, in which the light intensity is set as high and low levels in the morning and evening respectively; the road flatness is divided into "steeper" and "flat" levels; the road width calculated from the streetscape data is divided into "wider" and "narrower", and the green visibility is divided into "high" and "flat" levels. The road width calculated from the streetscape data is divided into "wider" and "narrower", and the green visibility is divided into "high", "medium", "low", "high" and "low", "Medium" and "Low" levels; the machine non-segregation is divided into "none", "drawing line segregation", "railing separation", "green belt separation " four levels; along the street building occupancy rate for "high", "medium", "low", "high", "medium", "low", "high", "medium", "low", "high", "medium", "low", "medium", "low", "medium", "low", "low". and "low" levels (Table 1). Then use SPSS analysis software to carry out combined orthogonal experimental design, after removing the duplicate tabs to get a total of 31 groups of options (including the absolute advantage tab). In order to create a virtual experience environment to restore the scenarios of the tabs, we combined sketch up modelling software and enscape rendering software to restore the scenarios in the options (Figure 4). The 31 sets of tabs were divided into three groups to create a questionnaire, in which the exact same tabs were used to create validity testing questions to ensure the credibility of the questionnaire data.

Baidu maps street view



Semantic segmentation



Tree Grass Sky Driveway Footway Building Streetlight

Fig.3 Streetview semantic segmentation

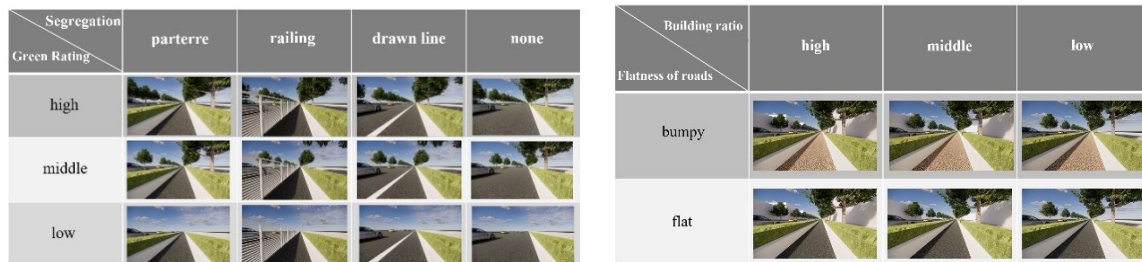


Fig.4 Virtual modelling scene

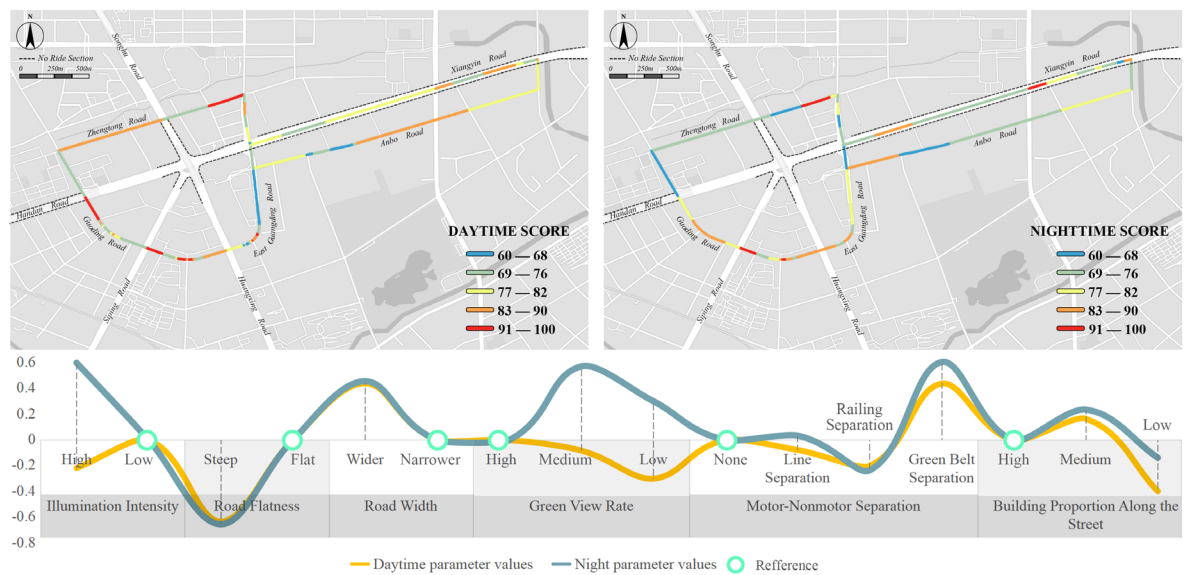
A total of 147 questionnaire results were collected, of which 13 invalid questionnaires were excluded. The preliminary results of the questionnaires showed that the research participants preferred cycling roads with low light intensity in the daytime, high light intensity at night, moderate road width, smooth roads, separated green belts, and high green visibility with a moderate percentage of buildings along the street. The Nlogit software was used for further in-depth analyses, and we used ML (mixed logit) modelling to determine the utility formula for the cycling environment and the corresponding weights for each factor. weights, and the results of the early and late analyses are shown in. Some of the factor weights have significant differences (Figure 5), which are mainly reflected in the daytime riding, cyclists preferred to ride in lower light, i.e., road sections with shade or shade, and the light intensity was negatively correlated with riding comfort; in the green visibility, the higher the green visibility, the higher the road comfort scores; for the nighttime riding, in the effect of the light intensity, cyclists preferred to ride in stronger light, i.e., road sections with sufficient illumination, and the light intensity was negatively correlated with the comfort. For night-time riding, in terms of the influence of light intensity, cyclists prefer to ride on road sections with strong light, i.e. with sufficient lighting, and light intensity is positively correlated with riding comfort; at this time, the blockage of green trees becomes an important obstacle to shade the light source, and for this reason, lower green visibility can improve the willingness to ride at night. In addition, some of the same sections showed differences in morning and evening suitability (Figure 5). For example, section 4 is considered to have good daytime riding conditions during the day due to sufficient green shade and open streets with a few buildings along the street, but at night it is considered to have poor nighttime riding conditions due to insufficient illumination and cold streets, which tends to lead to a lack of safety for cyclists. In summary, the morning and evening riding evaluations of the non-prohibited riding sections in Wujiaochang reveal the variability of riding conditions in different sections and at different times of the day, and that the branch road environment needs to be urgently improved in the face of the enormous pressure of assuming the increasing non-motorised

traffic flow in the Wujiaochang roundabout day by day.

This not only concerns the smoothness and safety of urban road traffic, but also involves the protection of and respect for the right of way of non-motorised vehicles. In the increasingly congested urban traffic, to ensure the right of passage of non-motorised vehicles is not only a test of the rationality of urban traffic planning, but also a basic guarantee for the safety of citizens travelling short distances.

Table 1 Elements of environmental impacts of bicycle travelling and their levels

Selected Factor	Level of Factor (Daytime)	Level of Factor (Night)	Questionnaire Parameter	Dummy Variable	Variable Name in Results
Illumination Intensity	High (4790.128476 - 14334.412009)	High (2.708757 - 7.801389)	1	1	LUX
	Low (923.002723 - 4790.128475)	Low (0.000000 - 2.708756)	2	0	BASE
Road Flatness	Steep (0.161551 - 0.442167)	Steep (1.016580 - 2.468278)	1	1	FLATNESS
	Flat (0.006556 - 0.161550)	Flat (0.000258 - 1.016579)	2	0	BASE
Road Width	Wider (0.215915 - 0.387865)	Wider (0.215915 - 0.387865)	1	1	WIDE
	Narrower (0.057253 - 0.215914)	Narrower (0.057253 - 0.215914)	2	0	BASE
Green View Rate	High (0.340360 - 0.697727)	High (0.340360 - 0.697727)	1	(0, 0)	BASE
	Medium (0.141980 - 0.340359)	Medium (0.141980 - 0.340359)	2	(1, 0)	GREEN1
	Low (0.008266 - 0.141979)	Low (0.008266 - 0.141979)	3	(0, 1)	GREEN2
Motor-Nonmotor Separation	None	None	1	(0, 0, 0)	BASE
	Line Separation	Line Separation	2	(1, 0, 0)	SEGMENT1
	Railing Separation	Railing Separation	3	(0, 1, 0)	SEGMENT2
	Green Belt Separation	Green Belt Separation	4	(0, 0, 1)	SEGMENT3
Building Proportion Along the Street	High (0.410132 - 0.555923)	High (0.410132 - 0.555923)	1	(0, 0)	BASE
	Medium (0.248302 - 0.410131)	Medium (0.248302 - 0.410131)	2	(1, 0)	BUILDING1
	Low (0.029036 - 0.248301)	Low (0.029036 - 0.248301)	3	(0, 1)	BUILDING2



	Illumination Intensity		Road Flatness		Road Width		Green View Rate			Motor-Nonmotor Separation				Building Proportion		
Variable Name In Results	HIGH	LOW	STEEP	FLAT	WIDER	NARROWER	HIGH	MEDIUM	LOW	NONE	LINE	RAILING	GREEN BELT	HIGH	MEDIUM	LOW
Daytime Parameter Values	-0.21843**	0	-0.63607***	0	0.44365***	0	0	-0.08208	-0.30088***	0	-0.07693	-0.20184*	0.44108***	0	0.16665	-0.39992***
Night Parameter Values	0.60397***	0	-0.65703***	0	0.45968***	0	0	0.57584***	0.30459**	0	0.03486	-0.23628*	0.60958***	0	0.23769**	-0.13823

Fig.5 Distribution of the main results of the model fitting and the composite evaluation scores of the no-riding road sections

3. Exploring the justice of the anti-riding policy

Historically, scholars have paid little attention to the issue of justice in the field of transport and have tended to deal with it only at the margins of their arguments.(Martens, Golub and Robinson, 2012) Cycling is a marginalised mode of transport compared to motorised transport, which receives more funding, attention and space. This marginalisation of cycling is widespread, even in cities that are otherwise hailed as successful cycling cities. (*The politics of cycling infrastructure. Spaces and (In)Equality: European Planning Studies: Vol 29 , No 1 - Get Access, no date*) This undoubtedly creates an inequitable distribution of transport and the rights of cyclists need to be safeguarded. When identifying transport inequity, there are usually two broad ways of understanding equity:social equity and spatial equity. (Lee, Sener and Jones, 2017) Therefore, this paper will explore the justice of the no-cycling policy from these two perspectives.

3.1 spatial justice

Spatial justice, initially has become a concept closely intertwined with efforts to make geographical distribution more equitable and sustainable. (Smith *et al.*, 2023)Achieving spatial justice would serve as a means to address the inequitable distribution of public and private goods, services, and resources within communities across the globe.

3.1.1 Space security

Historically, transport policies and research have predominantly concentrated on facilitating the safe movement of motorized vehicles, often neglecting the potential of eco-friendly travel options such as cycling. Currently, public policymakers are prioritizing the promotion of cycling as a sustainable mode of transport. Street safety stands out as a pivotal factor in encouraging the adoption of cycling as a green alternative, and it serves as a significant measure of spatial justice within urban transport systems. (Asadi-Shekari, Moeinaddini and Zaly Shah, 2013)

Cyclists and those considering cycling frequently point to personal safety concerns as a major deterrent. Studies have demonstrated that enhancements in infrastructure, such as the development of cycle lanes and tracks, have successfully led to an increased utilization of bicycles for transportation purposes (Buehler and Dill, 2016). With a focus on street safety and within the context of actual research findings, the cycling environment can be assessed from the following angles:

Bike Lanes: Dedicated bike lanes and their associated marking symbols have proven effective in reducing bicycle accidents.(Hamann and Peek-Asa, 2013) Bike lanes can be categorized into hard and soft segregation types: hard segregation involves the use of three-dimensional physical barriers (e.g., railings, green belts, cycle track corridors), while soft segregation refers to the use of two-dimensional delineation methods (painted lines, color differentiation, material division). The safety effectiveness of these segregation types requires periodic reassessment. Hard segregation is more effective in minimizing conflicts between motor vehicles and bicycles, offering better protection to cyclists. Conversely, soft segregation is more adept at reducing conflicts between bicycles and electric assistive vehicles, preventing overtaking conflicts, especially in narrower lanes.

Road Smoothness: Poor pavement quality is a leading cause of bicycle-related injuries.(Nyberg,

Björnstig and Bygren, 1996) Issues can be categorized as long-term, including structural problems like cracks and potholes, and short-term, such as slippery surfaces post-rainfall, icy conditions during winter, and roughness caused by sand, dust, or gravel.

Road Width: Wider lanes can reduce the incidence of same-direction skidding accidents and provide more room for evasive maneuvers. However, the safety benefits may be counterbalanced by the increased potential for higher speeds associated with wider lanes.(Schepers and Klein Wolt, 2012) Research indicates that, under normal conditions and speeds, bicyclists typically require about 1 meter of lane width to account for their natural weaving and to provide adequate space. A minimum width of 2 meters is recommended for one-way bike paths to ensure safe overtaking (Goodno *et al.*, 2013)

3.1.2 Spatial elasticity

Cycling is highly temporal in nature, with the frequency of cycling associated with a time period. (Loidl *et al.*, 2019) This change is reflected in the day, and is consistent with changes in people's daily commuting behaviour. It is divided into morning peak (mostly 7:00-9:00) and evening peak (mostly 17:00-19:00), the traffic demand of people going out to work and returning home from work, which is the time with the highest demand for cycling traffic, and an important time to test the carrying capacity of the supply and demand of cycling facilities.(吕 和 潘, 2018) Factors affecting cycling also change with the time of day, and the previous empirical study elaborated on the different roles of light and greenery on cycling willingness during daytime and nighttime. Therefore, the design of cycling infrastructure and cycling environments needs to take into account the specific needs of cyclists at different times of the day to achieve a dynamic balance that takes into account the chronological nature. At peak commuting times, the European policy of traffic calming can be used for specific streets, where cyclists have absolute traffic priority across the entire width of the street. Cars are usually allowed to drive on the street as well, but at a speed limit of 30 kilometres per hour, and cannot run into or otherwise interfere with cyclists. (Pucher and Buehler, 2008) In terms of the cycling environment, it is necessary to incorporate field research, light intensity measurements, appropriate roadside greenery, and ensure that street lighting is bright at night to provide bright and safe streets for cyclists.

3.1.3 Spatial empowerment

The concept of accessibility was defined as the opportunity to interact and overcome spatial separation. This definition underscores both the ease of interaction and the potential for interaction between different locations (Hansen, 1959). As the field of accessibility research progresses, it bifurcates into physical accessibility—encompassing factors such as right-of-way, road environment, and surrounding land use—and social accessibility, which considers variables like income, gender, and race.

In the realm of accessibility metrics, distance metrics, also recognized as connectivity metrics, represent the most straightforward category of location-based accessibility measurements. Relative accessibility is measured by the degree of connection between two places or points on the same surface. While the simplest form of this measure is the direct line between two points, infrastructure-based accessibility metrics such as average travel time and average speed between two locations can also serve as indicators of relative accessibility. In land use planning, distance measures are frequently employed as criteria for evaluating maximum travel time or distance to a location or transport infrastructure. When analyzing more than two potential destinations, contour measures become applicable. These metrics calculate the number of

opportunities accessible within a specified travel time, distance, or cost, or they measure the time or cost (fixed opportunity) required to reach a set number of opportunities (Kelobonye *et al.*, 2019).

Commuting time is inherently linked to the origins and destinations of travel. The area surrounding Wujiaochang is predominantly residential and educational, while the core of Wujiaochang constitutes an urban commercial and office hub. An isochronous circle analysis of these residential and educational areas reveals that the implementation of no-riding policies prevents cyclists from taking the most direct routes. As a result, they must expend additional time circumventing to reach their destinations, thereby increasing the cost of their travel time. This situation represents a spatial inequality when compared to the direct accessibility afforded to motorized traffic heading to the core area.

3.2 social justice

Social justice is characterized by the absence of systemic inequalities among different social groups, which involves reducing systematic discrimination against marginalized communities (Wiles and Kobayashi, 2009). Manifestations of social injustice are evident in the disparities across population groups, regions, and social identities regarding health and healthcare access, employment opportunities, educational resources, social support systems, and exposure to crime and environmental risks. These imbalances indicate the extent of exclusion experienced by certain groups, who are disproportionately affected in terms of social disadvantages such as income, wealth, opportunities, or health.

3.2.1 Affordability

The core goal of social justice in transportation is to "promote social and economic opportunity through equitable levels of affordable and reliable transportation choices based on the needs of the populations served, particularly those traditionally underserved". (Doran, El-Geneidy and Manaugh, 2021) Cycling is the most equitable in all modes of transport because it is affordable for almost everyone. However, the opportunity to be able to cycle is not always equitably distributed throughout society, which means that some people (e.g. low-income people, immigrant and racialised minority populations and groups such as women) do not have access to safe, reliable and convenient conditions for cycling. This is particularly problematic as many people belonging to such disadvantaged groups are more reliant on alternative modes of travel (e.g. cycling, walking and public transport) that are cheaper than cars. (Allen and Farber, 2019) Recently, scholars have begun to focus on methods and indicators used in economics to measure differences in income distribution, such as the Gini coefficient and Lorenz curve, the Theil index. (Cunha and Silva, 2023) When planners and policy makers fail to effectively integrate transport equity into planning and policy making, placing the interests of cyclists behind those of motorists can result in a disproportionate number of cyclists. drivers, it creates a disproportionate burden and fails to adequately address the needs of those who benefit the most. Empirical studies have found significant increases in cycling among commuters in disadvantaged areas in cities that have implemented bicycle initiative programmes. (Goodman *et al.*, 2013) It is clear that cycling among economically disadvantaged groups demand is huge and yet to be tapped.

3.2.2 Equitable distribution of health benefits

Given evidence that people of low socioeconomic status have disproportionately low access to active

transport infrastructure (Fletcher, Garasky and Nielsen, 2005) and tend to live in close proximity to major roads, they may be particularly susceptible to the risks of active transport, and their net health benefits may be correspondingly lower.(Day, 2006) In other words, while people of low socioeconomic status may derive health benefits from active transport, these benefits may be less than those of more advantaged populations. However, it is important to recognise that their proportional gains in health may still be higher, given their relatively high prevalence of chronic diseases associated with physical inactivity.(Lee, Sener and Jones, 2017)

The health benefits of cycling are, therefore, very significant for those with low socioeconomic levels. Wujiaochang is surrounded by many schools, residential areas, and business parks, and its population is predominantly composed of university students, elderly people living in older neighbourhoods, and white-collar workers who are new to the workforce, all with a generally lower economic level and a greater need for mobility in this area, making bicycles a natural and indispensable mode of transport for daily commuting. Through actual interviews and questionnaires, it has also been found that the enactment of the no-bicycle policy has significantly affected the convenience of their daily commuting, increasing the travel time and economic costs. In terms of social impact, this group of people generally lacks the opportunity to exercise due to time and financial constraints. The implementation of no-riding policies has impeded their access to cycling as a health-promoting mode of transport, potentially leading to a lack of health benefits such as weight loss and prevention of chronic diseases, including obesity, diabetes, and cardiovascular disease.(Barajas and Braun, 2021)

3.2.3 Participatory planning

Increasing public participation has significant attributes in mitigating public transport inequality. Effective public participation contributes to improved urban transport management, which will promote the procedural and normative development of public transport systems. (Fouracre, Sohail and Cavill, 2006) In view of the fact that the implementation of urban transport policy will affect the right to travel of all citizens, public opinions and suggestions should be heeded, and the actual demands of residents using different modes of transport should be fully reflected.

In cities, some disadvantaged groups may not be able to enjoy quality transport services due to economic and social status constraints. Through participatory planning, these groups can be directly involved in the decision-making process of cycling planning and express their needs and demands. This ensures that the planning scheme is more in line with their actual needs, improves the quality of their transport services, alleviates their travel pressure, and promotes social equity. An effective negotiation mechanism for policy settlement should also be established to reflect the positive impact of public contribution in rationalising public transport-related policies.

At the same time, a sound feedback mechanism can let the public know that their views have been taken into account, thus making them more enthusiastic to participate in policy decision-making. (Chen, F., & Ji, X. (2011). *Impacts analysis of transport equity under urban transport policy*. *Journal of Kunming University of Science and Technology*, 11(2), 7–10)Regardless of the form of participation channel, it is important to establish a sound disclosure, feedback, and evaluation mechanism. This allows the public to reflect on the actual benefits of enacted transport policies in a timely manner and dynamically adjust urban transport policies from the bottom up. The aim is to maximise the benefits to the largest number of groups and to fully take care of the interests of disadvantaged groups.

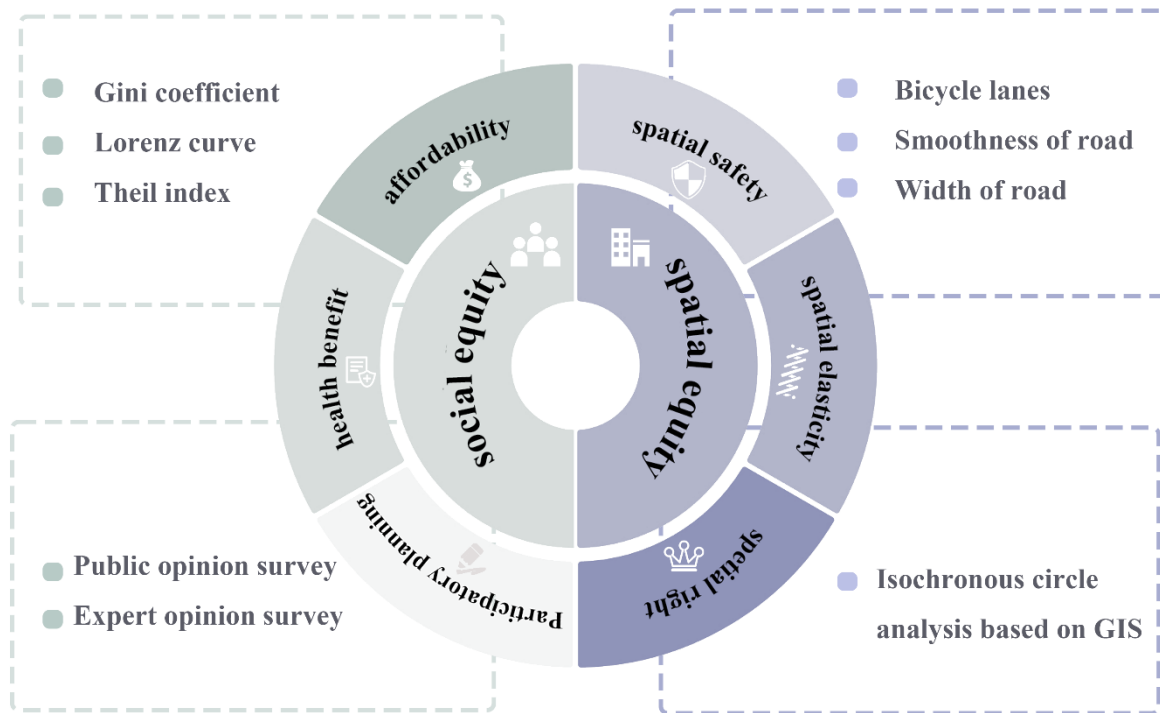


Fig.6 General overview of the equity discussion

4. Policy and Governance

4.1 Status of governance of the no-riding policy

In response to the growing contradiction between the ban on cycling, the Shanghai government has gradually discovered the irrationality of the ten-year ban on cycling policy in some areas, and has actively implemented a variety of measures in an attempt to restore the right of way of non-motorised vehicles on the banned road sections and the roads around them.

In Shanghai's recent urban planning and traffic management, the design and implementation of non-motorised transport policies and measures have paid particular attention to the effective management and optimisation of no-cycling zones. In Huangpu and Xuhui districts, two-way non-motorised traffic has been restored through road projects, solving the problem of one-way cycling bans on specific road sections. Innovative pilot projects in Yangpu District, such as the installation of non-motorised common boards, aim to restore non-riding traffic, while adding non-motorised lanes through road reconstruction, enhancing the travel environment around Handan Road and creating alternative cycling paths. Huangpu District has adopted the P+W model for takeaway deliveries on Huaihai Road and other prohibited areas to adapt to fast-paced urban life, while upgrading non-motorised paths in the South Riverside area to promote cycling for fitness and implement the city's concept of "slow-moving traffic". Xuhui District adjusted the non-motorised vehicle parking points to enhance the recognition of the no-riding space and reduce the possibility of cyclists mistakenly entering the no-riding section. In Jing'an District, in the implementation of the time-limited pedestrian street on University Road, the parking spaces on the surrounding roads were re-planned, and vehicles were guided to use the surrounding car parks, which not only ensured the access of transit vehicles,

but also reduced the traffic pressure and noise pollution on the pedestrian street. The optimisation of the crossing environment at intersections and the repositioning of the access function of auxiliary roads in Yangpu District, and the installation of additional safety isolation facilities, have improved the efficiency and safety of road access. In addition, Wujiaochang Street enhanced public awareness of the no-riding zone and reduced illegal riding behaviour by setting up non-motorised vehicle route maps and improving traffic signage recognition. These comprehensive policy measures reflect Shanghai's careful consideration of no-riding management.

However, judging from the frequency of violations on the banned road sections and the feedback from the public, although Shanghai has taken diversified measures to meet the challenges of the banned cycling policy, there are still problems such as the lack of preliminary surveys, strong limitations in coverage, and a lack of autonomy and completeness in implementation, which do not better guarantee the right of way for non-motorised vehicles, and the banned cycling policy is still of high limitations.

Tab2. Measures taken by districts in central Shanghai in response to the no-riding policy

(Source: Author based on official website <https://www.shanghai.gov.cn/draw>)

shore	Specific policy measures	times	typology
Shanghai-level project involving	Non-motorized Lane Restoration Project for Fuxing Middle Road (from Baoqing Road to South Xizang Road, spanning 3.3km) and Jianguo Road (West, Middle, and East sections, from South Wulumuqi Road to Zhaozhou Road, also 3.3km)	2018	Lift the one-way ban on cycling and restore two-way non-motorised traffic by, for example, compressing the width of non-motorised lanes.
Shanghai level project involving	Pilot the installation of additional non-motorised lanes through the installation of pedestrian and non-motorised common slabs at sections with large pavement widths along Military Road	2021	Manned and unmanned common boards, resumption of no-cycling traffic
Huangpu district, central Shanghai	The demand for takeaways from the Huaihai Road no-go zone is addressed.	2019	P(park)+W(walk) model
	In Huangpu District, the city's core with heavy traffic, "prohibited non-" road sections are being transformed into cycling-friendly paths around the South Riverside, aligning with local needs and advancing the city's "slow traffic" concept.	2023	Creating alternative cycling paths
	Thirty-three additional guidance signs have been installed on "no non-permit" roads in the central city, such as East Nanjing Road and Huaihai Road.	2018	Clarify the no-non-riding signs to increase the recognition of no-riding spaces
Xuhui district, central Shanghai	The non-motorised parking spots on the "no-non" roads have been shifted to the adjacent feeder roads. The use of peripheral roads to fill the parking function of the prohibited non-motorised road gaps, but also play a role in marking, to avoid cyclists because there are parking points and ride to the prohibited road sections.	2021	Modify the location of non-motorised parking spaces to increase the recognition of no-riding spaces
Jing'an district, central	For the launch of University Road's pedestrian street, parking on nearby Weicheng, Jinjia, and Weikang Roads was removed, directing vehicles to three nearby	2023	Creating alternative cycling paths; modifying the location of non-motorised parking spaces

Shanghai	car parks. This aimed to maintain traffic flow, minimize regional disruption, and reduce noise from the pedestrian street's opening.		
Yangpu district, central Shanghai	Road improvement of Guoding Road (Zhengli Road - Zhengtong Road)	2017	Road improvements to add railings to mechanically and non-motorised segregation, increase non-motorised lane widths, and continue to improve the non-motorised travel environment on the roads around Handan Road. Creating alternative cycling paths.
	Road Improvement of Zhengtong Road (Guoding Road - Guobin Road)	2018	Road improvements to add railings to mechanically and non-motorised segregation, increase non-motorised lane widths, and continue to improve the non-motorised travel environment on the roads around Handan Road. Creating alternative cycling paths
	Road Improvement of Guoji Road (Zhengdan East Road - Zhengtong Road) and Guobin Road (Handan Road - Zhengtong Road)	2019	
	Part of the green belt at the Handan Road Guoshun Road intersection was upgraded to widen the crossing facilities for pedestrians and non-motorised vehicles. In addition, also in the Handan Road Guoquan Road intersection on the east and west sides of each additional set of "pedestrian crossing" prompt system, and the intersection of the existing traffic signals, traffic signs and markings and guide islands to optimise the intersection of pedestrians and non-motorized vehicles within the scope of the right of way to enhance the efficiency of the road and safety.	2021	
	The access function of Handan Road (Guonian Road - Guoshun Road) was repositioned, two-way non-motorised lanes were set up, and safety isolation facilities were added to enhance the coherence and safety of the micro-circulation of the road network in this section.	2021	Resumption of No Ride Traffic
	The military road renovation plan addresses cyclist and pedestrian congestion at Zhoujiazui bus stops. Proposed changes include optimizing bus stop layouts, expanding waiting areas, and adding markings and guardrails to improve traffic flow and safety.	2022	Improved and optimised non-motorised vehicle-pedestrian conflicts on no-non road sections
	In order to make it easier for non-motorised vehicles to be informed of the distribution of permitted roads in the core area of Wujiaochang, Wujiaochang Street has set up non-motorised vehicle route maps around the core area of Wujiaochang.	2020	Clarify the no-non-riding signs to increase the recognition of no-riding spaces

4.2 Bicycle Policies in Other Key Chinese Cities

In response to the renaissance of cycling transport in China, Beijing formulated China's first specialised plan for cycling transport, the Beijing Bicycle and Pedestrian Transport Plan, back in 2016. The planning study focuses on right-of-way allocation in urban transport systems to achieve harmonious coexistence among multi-modal transport options and optimal resource allocation. Based on a comprehensive consideration of urban traffic demand, road capacity, and the interaction between different modes of transport, a human-centered right-of-way allocation strategy is proposed. This strategy adheres to the principle of social equity and aims to readjust and optimize road space usage by reducing the width of motorway lanes and eliminating on-street parking, thereby providing a safer and more convenient environment for bicycles and pedestrians. In addition, the plan's embedded parking space design and island bus stop layout not only solve the encroachment problems of motor vehicle parking and bus stops on cycle lanes but also enhance the comprehensive efficiency and safety of the transport system through refined right-of-way allocation. These measures reflect a respect for and balance of the rights and interests of different transport participants, offering innovative ideas for building an inclusive urban transport system.

Shenzhen, another international metropolis in China, is also a cycling-unfriendly city, similar to Shanghai. Beyond the same policy-based "no cycling" restrictions, Shenzhen's more prominent issue is the physical "no cycling" spaces, with a severe lack of non-motorized lanes and an increase in cut-off roads due to construction projects, all of which undermine cyclists' right of way in Shenzhen. With an influx of young people and a dramatic growth in cycling demand, Shenzhen enacted the Shenzhen Bicycle Transport Development Plan (2021-2035) in 2020. This comprehensive urban transport policy study is innovative in its integration of bicycle transport as a core component of the city's urban transport system, providing comprehensive policy support and an implementation framework for its development. To achieve the strategic goal of a 10-15% bicycle travel ratio by 2035, the plan adopts a systems engineering approach. It proposes five major development strategies—network reconfiguration, parking solutions, safety enhancement, cycling promotion, and policy synergy—through a comprehensive analysis of Shenzhen's geographic, social, and economic characteristics.

A breakthrough is made to construct a cross-regional bicycle skeleton network across the city, with priority given to repairing broken roads on high-demand cycling commute paths and gradually eliminating discontinuities in bicycle paths at rivers, mountains, large scenic spots, railways, high-speed roads, and large interchanges. This is achieved through the installation of additional bicycle towpaths, ramps, and bridges across rivers, or the reconstruction of current pedestrian bridges and cross-street tunnels. These strategies aim to optimize the layout of cycle paths, improve the safety and convenience of parking facilities, enhance cyclist safety and security, stimulate public demand for cycling, and ensure the consistency and synergy of policy measures. This approach differs from Shanghai's localized initiatives, which aim to mitigate the impact of the no-cycling policy by promoting the integration of cycling transport into the mainstream of urban transport with comprehensive policy support and infrastructure development.

4.3 Expand in detail on quality policies abroad

Policy portfolio theory suggests that the effects of policies are mutually reinforcing, and that the total effect of a policy package is greater than the sum of its individual effects.(Cejudo and Michel, 2021) To firmly establish a cycling-friendly city in Shanghai, it is necessary to adjust the existing no-riding policy and

introduce measures related to the physical environment of cycling, public policy, and socio-cultural aspects. The table below summarizes three types of measures widely adopted in cities with high cycling prevalence, such as the Netherlands, Denmark, and Germany.

In terms of the physical environment, both Amsterdam (population 735,000) and Copenhagen (population 504,000) have approximately 400 km of fully segregated cycle lanes and on-street cycle tracks.(Pucher and Buehler, 2008b) Even in smaller cities, the cycle network is well developed. Bridges also feature cycle lanes. Regarding bicycle parking, Amsterdam mandates a minimum number of bicycle parking spaces inside and near buildings, provides facilities around transport stops to prevent bicycle theft, and utilizes loading zones for trucks and daytime markets for nighttime bicycle parking. To prevent overcrowding, bicycle parking times are strictly regulated. The signage system is clear and unambiguous; cycle lanes are marked in red with a bicycle symbol, and neighborhoods use red brick surfaces to indicate bike territory, prompting cars to slow down. Traffic signals include those for bicycles, and signs for turning, going straight, and waiting are clearly marked to ensure safe cycling.

As a matter of public policy, the Netherlands has adopted a mixed-use land development model that provides housing for more people on the same land area and locates employment opportunities and amenities close to homes, reducing commuting times and increasing cycling possibilities. In the Netherlands, Denmark, and Germany, most residential streets are traffic-calmed, with a legal speed limit of 30 km/h and often no through traffic, providing a comfortable and safe environment for cyclists. Traffic laws in these countries offer special provisions to protect cyclists' vulnerable position in accidents with motor vehicles (German Federal Ministry of Transport, 2006), encouraging motorists to be more cautious and proactive in avoiding collisions.

Socio-culturally, the Netherlands collaborates with local universities to improve cycling policies and promote a cycling culture through research projects and community cooperation. Bicycle festivals and car-free days are organized to highlight the benefits of cycling, fostering a social climate that supports cycling. The country also actively seeks public feedback on transport policies, regularly surveys the public's wishes, and holds expert symposiums to provide real-time, efficient feedback on the shortcomings of existing policies.

Tab3. Summary of foreign cycling policies

form	initiative	Representative policy papers	The Shanghai Borrowing Approach
Physical environment	cycle track	City of Amsterdam. The Amsterdam Bicycle Policy. Amsterdam: Dienst InfrastructuurVerkeer en Vervoer, 2003a.	Fill and upgrade urban cycle lanes to form a complete and continuous cycling network.
	Bicycle parking systems	City of Berlin. Bauordnung fuer Berlin. City Construction Code [S]. Berlin. Senatsverwaltung fuer Stadtentwicklung, 2005.	Provision of a prescribed minimum number of bicycle parking spaces within and near buildings; provision of bicycle parking facilities around transport stops (e.g. bicycle management stations, bicycle racks and other parking facilities to prevent bicycle theft).
	Bicycle signage	City of Amsterdam. Choosing for the Cyclist: Bicycle Program 2007-	Use eye-catching colours and vivid patterns to mark bicycle lanes,

	system	2010[M]. Amsterdam: Gemeente Amsterdam, 2007.	bicycle parking points and other related facilities.
Public policy	Traffic calming	City of Berlin. Fahrradverkehr [Bicycling Transportation in Berlin] [M]. Berlin: Senatsverwaltung fuer Stadtentwicklung, 2007 [2007-11-30]. http://www.stadtentwicklung.berlin.de/verkehr/radverkehr/index.shtml .	The streets around the no-riding zone are uniformly controlled for traffic, with the legal speed limit reduced to 30 kilometres per hour and any crossing vehicles frequently prohibited.
	Traffic Regulations	German Federal Ministry of Transport. Strassenverkehrsordnung [German Traffic Laws]. Berlin: German Federal Ministry of Transport, 2006 [2007-11-30]. http://www.bmvbs.de/Anlage/original_987610/StrassenverkehrsOrdnung.pdf .	Take care of the vulnerable position of cyclists and increase the penalties for misbehaviour by motor vehicle traffic endangering cyclists.
	Functionally mixed land development	City of Amsterdam. The Amsterdam Bicycle Policy. Amsterdam: Dienst InfrastructuurVerkeer en Vervoer, 2003a.	Enrich the types of functions within the site and reduce the distances people travel.
Social culture	Focus on partnerships with local universities	Netherlands Ministry of Transport. Passenger Kilometers of Cycling and Fatality Rates: Data Gathered Directly from the Ministry of Transport. Rotterdam: Ministry of Transport, Public Works, and Water Management. 2007.	Collaboration with relevant professions at Fudan University, Tongji University and Jiaotong University in Shanghai has led to their inclusion in the establishment of cycling-friendly cities.
	Cycling Culture	Netherlands Ministry of Transport. Cycling in the Netherlands [R]. Rotterdam: Ministry of Transport, Public Works, and Water Management. 2006.	Organise thematic cycling activities to promote a cycling atmosphere for all; enhance the coverage of cycling-related benefits to strengthen the community's willingness to cycle
	Participatory planning	City of Berlin. Fahrradverkehr [Bicycling Transportation in Berlin] [M]. Berlin: Senatsverwaltung fuer Stadtentwicklung, 2007 [2007-11-30]. http://www.stadtentwicklung.berlin.de/verkehr/radverkehr/index.shtml .	To conduct regular surveys on cyclists' satisfaction with the facilities and activities, and to collect opinions and suggestions for further improvement. Organise cycling conferences to provide a platform for groups such as cycling companies, the cycling industry, city authorities, research institutes, universities, cycling experts, citizens' representatives, etc. to exchange views.

5 Conclusion and Discussion

This paper examines the no-riding policy in the Wujiaochang roundabout area of Shanghai's Yangpu District. Through a questionnaire, we assessed the public's understanding of the policy and their views on its implementation across different road sections. The findings indicate that the policy is not widely supported and that a uniform approach does not meet the diverse needs of the roundabout. To evaluate the policy's impact, we assessed the cycling environment of adjacent side roads, using indicators such as green visibility, lack of mechanical segmentation, building density along the street, and road width, derived from Baidu Map Street View, machine learning, and semantic segmentation. Road flatness and illuminance were measured using on-site cycling and a sensor logger.

Cyclists prefer roads with low light intensity during the day, high light intensity at night, moderate road width, smoothness, green belt separation, high green visibility, and a moderate building density along the street. These preferences should guide future cycling policy formulation. The cycling ban policy adversely affects cyclists' spatial safety, resilience, and rights, potentially depriving disadvantaged groups of opportunities to improve their employment and health. It also highlights the absence of participatory planning in current transport policy, which does not adequately represent public demands.

Scraping street map data is a universally applicable, convenient, efficient, and accurate method for objectively evaluating the existing cycling environment based on indicators such as green visibility, non-mechanized segmentation methods, and building density along the street. Using a sensor logger to collect cycling data in the field simplifies the analysis of road flatness and illumination, providing first-hand, accurate, and intuitive data. This combination is highly operational and suitable for most areas.

Our study concludes that the cycling environment around the Wujiaochang roundabout requires improvement, with enhancements needed in green visibility, non-mechanized segmentation, building density, road smoothness, and light intensity according to the time of day. The no-riding policy should be adjusted to current traffic demands, with the possibility of lifting restrictions in some sections or improving the environment of surrounding branch roads.

A limitation of this study is its focus on spatial physical indicators of cycling justice, neglecting social indicators such as socio-economic status, cultural background, ethnicity, and gender. Future research should aim to include these social indicators to more comprehensively evaluate the fairness and social inclusiveness of the cycling environment, leading to a deeper understanding of injustices and more targeted policy and planning solutions.

Based on the experimental results and fairness analyses of the Pentagon, we get the idea that "different zones have different speeds"

Urban planning must acknowledge the inherent complexity and the impossibility of absolute equity. The goal should be to strategically create zones where cycling can thrive with a sense of relative equity. Recognizing that motor vehicles will naturally dominate the city's major highway and arterial road systems, it's essential to envision a multi-speed urban fabric where different modes of transport are prioritized according to the context.

In the smaller, more intimate neighborhoods, non-motorised transport should be given precedence for short-distance travel, fostering a slower, more community-oriented pace. Conversely, in expansive public spaces such as Shanghai's riverside areas, the cultural significance of cycling can be leveraged to create a shared, vibrant space where pedestrians and cyclists coexist, each moving at their own comfortable pace without the dominance of motor vehicles.

The challenge for future policymakers and urban designers lies in crafting a nuanced transportation policy that embraces this concept of 'different speeds for different regions.' This approach should be central to the evolution of Shanghai's urban planning, particularly as it pertains to the development of cycling infrastructure within rejuvenated neighborhoods. By doing so, Shanghai can become a model for a bicycle-friendly city that not only accommodates diverse transportation needs but also enriches the cultural tapestry of its communities.

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